

Exploratory Example — FMSM Public v1.0

This document presents qualitative examples of model behavior with and/or without FMSM Public v1.0. Outputs are reproduced for illustration and may contain errors, unsupported claims, or residual reasoning failures.

These examples are not a benchmark, controlled experiment, or demonstration of superiority. They are provided to illustrate observable differences in analytical behavior and the continuing influence of the underlying model.

Model: GPT-5.6 Sol / Gemma 4 12B QAT

Condition: FMSM Public v1.0 / Vanilla

Run: Single run

Léa was born into a family of small business owners who highly valued financial independence and a wariness of debt.

At 18, she entered a prestigious business school where her peer group, by contrast, prized risk-taking, startups, and external financing.

At 25, she founded a company. She initially refused all outside funding but gradually accepted venture capital after two failed attempts at self-funded growth.

At 35, having become an investor, she encouraged her portfolio companies to raise capital quickly, yet personally maintained very low debt levels and a substantial cash reserve.

At 42, following a financial crisis that caused two highly leveraged companies in her portfolio to go bankrupt, Léa began requiring the new companies she funded to maintain twelve months of available cash, while still recommending that they raise external capital to finance their growth.

ChatGPT 5.6 Sol

Final Report — FMSM Emergent Causalities Analysis

Executive Assessment

Léa's trajectory shows a **selective adaptation of financial behavior rather than a simple reversal of beliefs.**

The strongest conclusion supported by the premise is that her behavior differentiates increasingly between two financial functions:

- **external capital as a mechanism for financing growth**, which she progressively accepts and later recommends;
- **financial resilience through low leverage and liquidity**, which remains visible in her personal behavior and is later imposed on portfolio companies after observed bankruptcies.

This distinction is directly compatible with all observed stages of the trajectory. What cannot be established from the premise alone is **why** Léa preserves some practices while changing others. Persistence of early family socialization, rational learning from experience, risk aversion, path dependence, compartmentalization, or combinations of these mechanisms remain competing explanations.

The crisis at age 42 does not return Léa to her initial rejection of external financing. Instead, it produces a new configuration: **external financing is retained, while vulnerability to leverage and insufficient liquidity is constrained.**

Overall epistemic assessment: the behavioral trajectory is strongly established at D0; the mechanisms explaining it are mostly D1; psychological or motivational explanations remain D1–D2 hypotheses.

STRUCTURAL ANALYSIS

1. Established Facts

D0 — Directly established by the premise

Léa is raised in a family of small business owners where two financial principles are highly valued:

1. financial independence;
2. wariness of debt.

At 18, she enters a prestigious business school whose peer environment values:

- risk-taking;
- startups;
- external financing.

At 25, when founding her own company, Léa initially refuses all outside financing.

She subsequently experiences two failed attempts to achieve growth through self-financing.

After those failures, she progressively accepts venture capital.

At 35, as an investor, she exhibits two simultaneous patterns:

For portfolio companies: she encourages rapid capital raising.

For herself: she maintains very low debt and substantial cash reserves.

At 42, a financial crisis results in the bankruptcy of two highly leveraged portfolio companies.

Following this event, she requires newly funded companies to maintain twelve months of available cash reserves.

She nevertheless continues recommending external capital raising to finance growth.

These elements establish a behavioral trajectory. They do **not**, by themselves, establish Léa's internal reasoning or motivations.

2. Dominant Mechanisms

Mechanism 1 — Exposure to competing financial norms

Epistemic status: D1 — strong inference

Léa is sequentially exposed to two contrasting financial environments.

Her family environment favors:

autonomy + debt caution.

Her later peer environment favors:

risk acceptance + startups + external financing.

The premise establishes exposure to these norms but does not establish their respective causal strength.

A plausible mechanism is therefore **normative competition**: Léa acquires access to financial models that differ from those dominant during her childhood.

It would be excessive, however, to conclude that business school *caused* her later acceptance of venture capital. Her initial refusal of outside financing at 25 indicates that exposure alone did not immediately produce behavioral adoption.

Mechanism 2 — Persistence of previously acquired financial behavior

Epistemic status: D1

Her initial refusal of outside financing is behaviorally consistent with the financial norms of her family environment.

Her low personal debt and substantial liquidity at 35 are also compatible with those norms.

This supports—but does not prove—a persistence mechanism in which earlier financial practices remain influential despite exposure to competing models.

Alternative explanations remain possible. Low personal leverage at 35 could result from her current financial circumstances, rational portfolio management, personal preferences acquired later, or factors absent from the premise.

Therefore:

Observed continuity ≠ demonstrated psychological inheritance.

Mechanism 3 — Learning through outcome feedback

Epistemic status: D1 — strongly supported

Two important behavioral changes occur after negative outcomes.

First sequence:

self-funded growth → two failures → progressive acceptance of venture capital

Second sequence:

highly leveraged portfolio companies → crisis → two bankruptcies → mandatory liquidity reserves

The temporal structure strongly supports an experiential-learning mechanism.

The precise causal interpretation remains somewhat uncertain because the premise does not state that Léa explicitly attributed either decision to those failures.

Nevertheless, the repeated pattern makes feedback-based adaptation one of the strongest inferred mechanisms in the scenario.

Mechanism 4 — Selective rather than global updating

Epistemic status: D1 — strong inference

Léa does not appear to replace one financial doctrine with another wholesale.

After self-funded growth fails, she accepts external capital.

After leveraged companies fail, she does **not** abandon external capital.

Instead, she adds a liquidity requirement.

The resulting configuration is approximately:

Growth financing: external capital remains acceptable.

Balance-sheet resilience: high liquidity becomes mandatory.

This suggests that Léa updates specific components of her financial model independently rather than treating “external financing,” “debt,” “capital raising,” and “liquidity” as interchangeable concepts.

This distinction is important because venture capital is not debt. Therefore, her acceptance of venture capital does not necessarily contradict continued wariness of indebtedness.

Mechanism 5 — Asymmetric risk treatment

Epistemic status: D1–D2

At 35, Léa recommends rapid fundraising for portfolio companies while maintaining low personal debt and substantial liquidity.

This could reflect different risk models for different contexts:

- personal wealth preservation;
- entrepreneurial growth;
- portfolio diversification;
- investor versus founder exposure.

It could also reflect psychological compartmentalization or differing tolerance for personal versus externally distributed risk.

The premise cannot discriminate among these possibilities.

3. Identified Strategies

Strategy A — Capital-assisted growth

D0 behavior / D1 strategic characterization

After her failed self-funded growth attempts, Léa accepts venture capital and later recommends external fundraising to portfolio companies.

A reasonable strategic interpretation is:

use external capital to overcome growth constraints.

The objective is inferred rather than explicitly stated, although the premise directly connects external capital to financing growth at age 42.

Strategy B — Liquidity-based resilience

D0 behavior / D1 strategic interpretation

Léa maintains substantial personal liquidity and later requires twelve months of cash reserves from funded companies.

The observable policy is established.

Its interpretation as a resilience strategy is strongly supported because it follows bankruptcies during a financial crisis, but Léa's precise reasoning is not stated.

Strategy C — Separation of growth capital from financial fragility

D1

By age 42, Léa simultaneously recommends:

raise external capital for growth

and:

maintain twelve months of liquidity.

This combination suggests a strategy in which access to external financing is preserved while exposure to financial distress is reduced.

Importantly, this is not necessarily contradictory.

External equity capital, leverage, and liquidity are different variables.

The mature policy therefore appears less like a compromise between two incompatible doctrines than a **decomposition of financial risk into separate dimensions**.

4. Stated or Candidate Motivations

No direct statement of Léa's internal motivations is provided.

Consequently, no motivation can be classified as established.

Several candidate motivations remain compatible with the facts.

Candidate motivation 1 — Preserve autonomy and security

D1–D2

Supported by her persistent low personal debt and high liquidity.

Potentially compatible with early family norms.

Not sufficient to explain why she actively recommends external capital raising.

Candidate motivation 2 — Maximize growth opportunities

D1

Supported by her progressive acceptance of venture capital after failed self-funded growth and her later recommendations to portfolio companies.

Does not independently explain her liquidity requirements.

Candidate motivation 3 — Reduce downside exposure without sacrificing growth

D1

This is particularly compatible with her age-42 policy.

External capital remains available for expansion while cash reserves provide a buffer against adverse conditions.

Candidate motivation 4 — Apply lessons from personally observed failures

D1

The timing supports this interpretation strongly.

However, the premise does not reveal whether Léa interprets the failures as evidence about leverage, liquidity, macroeconomic shocks, management quality, or some combination thereof.

HYPOTHESIS & RISK ANALYSIS

Competing Hypothesis A — Persistent early financial socialization

Claim: Léa's family environment creates a durable preference for financial independence, low debt, and liquidity, which remains present even after she adopts external financing.

Supporting evidence: Her initial rejection of outside financing; persistent low personal debt; substantial personal liquidity; later requirement for corporate cash reserves.

Counter-evidence: She eventually accepts venture capital and actively encourages external fundraising.

Missing information: Léa's own explanations, behavior before age 25, attitudes toward debt versus equity, comparison with peers exposed to similar environments.

Epistemic status: D1–D2, plausible but unconfirmed.

Competing Hypothesis B — Primarily experiential updating

Claim: Léa's trajectory is driven mainly by observed outcomes rather than durable childhood socialization.

Supporting evidence: Major behavioral changes follow concrete failures twice.

Two failed self-funded growth attempts precede acceptance of venture capital.

Two leveraged-company bankruptcies precede mandatory liquidity reserves.

Counter-evidence: Her low personal leverage and high liquidity persist even during the period in which she recommends aggressive fundraising to others.

Missing information: Whether these personal practices existed continuously, whether they changed after specific events, and Léa's explicit reasoning.

Epistemic status: D1, strongly compatible with the trajectory.

Competing Hypothesis C — Context-dependent financial optimization

Claim: Léa does not hold contradictory beliefs; she applies different financial rules to different economic problems.

Growth requires capital.

Personal security favors low leverage.

Corporate resilience requires liquidity.

Supporting evidence: Her behavior becomes increasingly differentiated by context rather than converging on either complete financial independence or unrestricted risk-taking.

Counter-evidence: The premise provides insufficient information about whether these distinctions are deliberate.

Missing information: Investment criteria, portfolio structure, risk models, target growth rates, debt/equity distinctions in her decision-making.

Epistemic status: D1, strongly compatible but not directly established.

Competing Hypothesis D — Personal risk aversion combined with externalized risk tolerance

Claim: Léa tolerates greater financial risk when borne by companies or investors than when borne personally.

Supporting evidence: Low personal debt and high personal liquidity coexist with recommendations for rapid external fundraising.

Counter-evidence: Raising equity does not necessarily increase company financial risk in the same manner as leverage. Her later liquidity requirements also demonstrate concern for portfolio-company resilience.

Missing information: Funding instruments, ownership exposure, personal guarantees, portfolio diversification, and Léa's actual risk preferences.

Epistemic status: D2; possible, but substantially less established than A–C.

Vulnerabilities & Mismatches

The most important analytical risk is treating all forms of financing as one variable.

They are not.

Debt ≠ venture capital ≠ external financing ≠ leverage.

Consequently, the apparent contradiction between Léa's childhood wariness of debt and her later acceptance of venture capital may be much weaker than it initially appears.

A second risk is interpreting chronological succession as sufficient proof of causality. The two failure → behavioral-change sequences are highly suggestive, but other variables may have changed simultaneously.

A third risk is psychological overinterpretation. The premise establishes behavior much more strongly than it establishes beliefs.

Effectiveness Levers

The mature strategy at 42 could be effective if several conditions hold:

- external capital genuinely alleviates growth constraints;
- the twelve-month liquidity requirement meaningfully reduces distress risk;
- maintaining liquidity does not impose excessive opportunity costs;
- companies do not substitute other forms of hidden risk for leverage;
- capital remains accessible when needed.

These are conditional mechanisms, not established outcomes.

UNCERTAINTIES & LIMITATIONS

What Is Unknown

The premise does not establish:

Léa's explicit beliefs at each stage; whether she distinguishes debt from equity conceptually; the causes of the two failed growth attempts; the precise causes of the two bankruptcies; whether leverage itself caused those bankruptcies; whether twelve months of liquidity would have prevented them; how her personal wealth is structured; or whether her investment strategy changed in other ways after the crisis.

These missing variables prevent strong psychological attribution.

Major Counter-evidence

The strongest counter-evidence against a simple "childhood beliefs determine adulthood behavior" explanation is Léa's substantial adaptation to external financing.

The strongest counter-evidence against a simple “business school replaced childhood beliefs” explanation is her initial refusal of funding and continued conservative personal balance sheet.

The strongest counter-evidence against a simple “crisis made her conservative again” explanation is that she continues recommending external capital after the crisis.

No single-environment explanation adequately captures the full D0 trajectory.

Sufficiency Test

Consider the explanation:

“Léa behaves this way because of her family upbringing.”

It explains reasonably well her initial reluctance toward outside funding and potentially her persistent personal conservatism.

It does **not** sufficiently explain her later acceptance and promotion of external capital.

Now consider:

“Business school changed her attitude toward finance.”

This may contribute to her exposure to alternative financial models, but it does not explain why she initially rejects outside funding seven years later or why conservative personal financial behavior persists.

Finally:

“Experience taught her what works.”

This explains the timing of both major adaptations better than either socialization-only explanation. But it still does not fully explain why some practices persist across contexts.

The available evidence therefore favors no monocausal explanation.

A multi-mechanism model is sufficient with fewer unexplained observations:

prior dispositions/norms + exposure to alternatives + outcome feedback + context-dependent adaptation.

The first component remains less firmly established than the latter behavioral sequence.

Removal Test

If all unestablished motivational and psychological hypotheses are removed, the following conclusions survive:

Léa begins with behavior consistent with financial conservatism.

She encounters a social environment favoring substantially different financial practices.

She initially continues to avoid external funding.

After two unsuccessful self-funded growth attempts, she progressively adopts venture capital.

She later recommends external capital to companies while personally retaining low debt and high liquidity.

After two highly leveraged portfolio companies fail during a crisis, she introduces a substantial liquidity requirement.

She does not abandon external capital as a growth mechanism.

Therefore, the strongest hypothesis-independent conclusion is:

Léa's observed financial behavior evolves through selective modification rather than complete replacement: practices associated with growth financing change, while practices associated with liquidity and limited leverage persist or are subsequently reinforced.

This conclusion requires minimal assumptions beyond D0.

CAUSAL SYNTHESIS

Trajectory

The observed trajectory can be represented conservatively as:

Family financial norms

↓

wariness of debt / financial independence valued

↓

Exposure to competing peer norms at 18

↓

external financing and risk-taking become available alternative models

↓

Company founded at 25

↓

outside financing initially rejected

↓

two unsuccessful self-funded growth attempts

↓

external capital progressively accepted

↓

Investor by 35

↓

external fundraising recommended to companies

+

low personal debt / high personal liquidity retained

↓

Financial crisis before 42

↓

two highly leveraged portfolio companies fail

↓

liquidity requirement introduced

+

external growth financing retained

↓

Hybrid financial configuration at 42

Conservative Synthesis Compilation

The evidence does **not** support the simple conclusion that Léa abandoned her family's financial model for that of her business-school peers, nor that she later returned to her childhood model.

Her observable trajectory is better described as **cumulative and selective adaptation**.

She initially behaves in a manner compatible with the financial norms of her upbringing despite later exposure to a competing model. Following unsuccessful self-funded growth, she progressively accepts external capital. Yet conservative personal debt and liquidity practices remain observable. Following later bankruptcies among highly leveraged portfolio companies, she strengthens liquidity requirements without abandoning external financing for growth.

The strongest inferred mechanism is **experience-dependent updating**, because two major policy changes follow adverse outcomes. Persistence of early socialization is also compatible with the trajectory but cannot be independently established from behavior alone.

The case therefore supports a distinction between **replacement of a belief system** and **reconfiguration of individual components within it**.

At age 42, Léa's behavior is consistent with a model in which:

capital can be used aggressively as a growth resource while liquidity and limited leverage are used defensively as resilience mechanisms.

Whether this synthesis reflects deliberate financial theory, persistent childhood socialization, learned risk management, psychological compartmentalization, or some combination remains unresolved.

Decision-maker assessment: High confidence in the behavioral trajectory; moderate confidence in selective/experiential updating as the dominant mechanism; low-to-moderate confidence in any specific psychological or motivational explanation.

Gemma4 12b QAT

This analysis follows the FMSM Protocol to deconstruct the trajectory of Léa's financial behavior and investment strategies.

STRUCTURAL ANALYSIS

Established Facts (D0):

Origin: Family of small traders; values: financial autonomy, distrust of debt.

Education: Attended a "grande école"; social environment values: risk-taking, startups, external funding.

Early Career (Age 25): Founded a company; initially refused external funding; accepted Venture Capital (VC) only after two failures in self-funded growth.

Mid-Career (Age 35): Became an investor; encourages portfolio companies to raise capital rapidly; maintains personally low debt and high liquidity.

Recent Event (Age 42): Two highly indebted portfolio companies failed during a financial crisis.

Current Policy: Mandates 12 months of cash reserves for new investments; continues to recommend external capital for growth.

Dominant Mechanisms:

Cognitive Dissonance/Synthesis: The interplay between "autonomy/debt-aversion" (family) and "leverage/risk" (school).

Adaptive Learning Loop: Transition from self-funding

→

→ external funding

→

→ regulated external funding based on empirical failure.

Risk Asymmetry Management: Distinguishing between growth levers (external capital) and survival buffers (liquidity reserves).

Identified Strategies:

Growth Strategy: Utilization of external capital to accelerate scaling (accelerated by the "grande école" influence).

Defensive Strategy: Mandatory liquidity buffers to decouple operational growth from immediate insolvency risk (influenced by family background and recent portfolio losses).

Stated or Candidate Motivations:

Stated/Observable: Seeking rapid growth through external capital; seeking to avoid bankruptcy through liquidity mandates.

Candidate (Inferred):

Hypothesis 1: Psychological Synthesis. Seeking a "middle way" that satisfies both the family's safety requirements and the peer group's growth ambitions.

Hypothesis 2: Risk Mitigation. A specific reaction to the "tail risk" of systemic financial crises (the age 42 event).

HYPOTHESIS & RISK ANALYSIS

Competing Hypotheses:

Hypothesis A: The Synthesis Model (Integrated Identity)

Supporting evidence: Her behavior at age 35 (pushing VC while keeping personal liquidity) suggests she has integrated both influences.

Counter-evidence: None currently available.

Missing information: Her internal reflections on why she separates personal and corporate debt structures.

Epistemic status: Inference (D1).

Hypothesis B: The Trauma/Scarring Model (Reactive Adaptation)

Supporting evidence: The specific shift at age 42 follows a direct, painful loss of capital due to debt. The "12-month rule" is a direct reaction to a specific failure.

Counter-evidence: If the rule were purely reactive, one might expect her to stop recommending external capital entirely, but she continues to do so.

Missing information: Whether she sought outside counsel or analyzed the specific causes of the failures.

Epistemic status: Hypothesis (D2).

Vulnerabilities & Mismatches:

The "Growth-Security" Paradox: There is a potential friction between "rapid capital raising" (which often involves debt or equity dilution that can strain liquidity) and the "12-month reserve" mandate. If growth is too aggressive, the 12-month reserve may become a "dead" asset that slows down reinvestment.

Personal vs. Professional Bifurcation: She maintains a high degree of personal liquidity while encouraging others to take risk. This suggests a high degree of "agency distance"—she views the portfolio as a laboratory for growth but her personal life as a fortress of security.

Effectiveness Levers:

Liquidity Buffer as a "Circuit Breaker": The 12-month rule acts as a stabilizer against market volatility without preventing the use of leverage for expansion.

UNCERTAINTIES & LIMITATIONS

What Is Unknown: The specific type of external capital she recommends (Debt vs. Equity). If she recommends debt, the 12-month reserve is the primary safety mechanism; if she recommends equity, the reserve is a secondary buffer.

Major Counter-evidence: If the two companies at age 42 failed due to poor management rather than high debt, her new rule might be a misaligned solution to the actual problem (i.e., she is treating a management problem with a liquidity fix).

CONSERVATIVE SYNTHESIS

The trajectory shows a transition from Debt-Aversion (Family)

→

→ Leverage-Acceptance (Education)

→

→ Regulated Leverage (Experience). Léa has moved from a binary view of finance to a nuanced, multi-layered approach. She treats "growth" and "survival" as two different variables: growth is fueled by external capital (the "Grande École" influence), while survival is ensured by liquidity buffers (the "Family" influence). The age 42 event served as a catalyst to formalize the "survival" side of her strategy into a mandatory policy.

FINAL REPORT

To: Decision-Maker

Subject: Causal Analysis of Investment Behavior – Léa

Summary of Findings:

Léa's investment profile is characterized by a hybridized risk model. She does not choose between her family's values (security/autonomy) and her peers' values (risk/leverage); instead, she has structured her strategy to accommodate both simultaneously.

Key Observations:

Facts: Léa has moved from 100% self-funding to advocating for external capital, but she has introduced a non-negotiable 12-month liquidity buffer following a portfolio failure at age 42.

Interpretation of Strategy: She employs a "Buffered Growth" model. She views external capital as an accelerator but views liquidity as an insurance policy. Her personal financial behavior (low debt/high liquidity) reflects a permanent internalized "safety floor" derived from her upbringing.

Stated vs. Inferred Motivations: While her stated goal is to protect companies from failure, the evidence suggests a deeper synthesis of identities. She seeks the upside of the startup world while maintaining the defensive posture of her family background.

Potential Risks: The primary risk is a mismatch in "velocity." If her portfolio companies need to move faster than the 12-month liquidity buffer allows, they may lose competitive advantages to less-constrained rivals.

Conclusion:

Léa is a "Risk-Aware Growth" investor. Her decisions are not a rejection of her past, but a chronological layering of experiences where each new life stage has added a new constraint to the previous one.

Gemm4 12b QAT _ Vanilla analysis

This narrative provides a compelling study of psychological synthesis—the process by which an individual balances two opposing ideologies to create a unique, nuanced philosophy.

Léa's journey can be broken down into four distinct developmental stages:

1. The Conflict of Identity (The Foundation)

Léa began with a "Fortress Mentality" (inherited from her family). This mindset prioritizes survival, autonomy, and the avoidance of obligation (debt). However, her environment at age 18 introduced the "Expansionist Mentality" (her peers). This mindset prioritizes velocity, scale, and the use of "other people's money" to accelerate progress.

2. The Pragmatic Adaptation (Age 25)

At 25, Léa experienced a "failure of idealism." By trying to stick strictly to her family's values (self-funding), she hit a growth ceiling. Her decision to accept venture capital wasn't an abandonment of her roots, but a pragmatic concession. She learned that while debt is a risk, stagnation is also a risk.

3. The Bifurcated Identity (Age 35)

By 35, Léa developed a "dual-track" personality. This is a common trait among successful leaders who bridge different worlds:

Externally (The Architect): She acts as a catalyst for growth. She understands the "Peer" logic—that capital is a tool for speed.

Internally (The Sentinel): She maintains the "Family" logic. She keeps her personal finances as a fortress, ensuring that even if her external ventures fail, her personal foundation remains untouched.

4. The Synthesis of Wisdom (Age 42)

The financial crisis served as the catalyst for integration. Before the crisis, she saw "growth" and "safety" as somewhat separate entities. After watching leveraged companies collapse, she codified a new philosophy: Growth with a Margin of Safety.

Her new requirement—raise external capital but maintain 12 months of cash—is the ultimate synthesis of her life's influences:

From her peers: The wisdom that growth requires external resources.

From her parents: The wisdom that debt without a buffer is a vulnerability.

Conclusion: The "Léa Model"

Léa has evolved from a person who was torn between two styles into a leader who perfected a third. She no longer chooses between being a "conservative" or a "risk-taker." Instead, she practices "Calculated Expansion."

She understands that the most sustainable way to build a business is to use the tools of the aggressive (capital) while maintaining the soul of the conservative (liquidity). She has moved from reaction (avoiding debt) to strategy (managing risk).